

CS 506 lab 11 questions

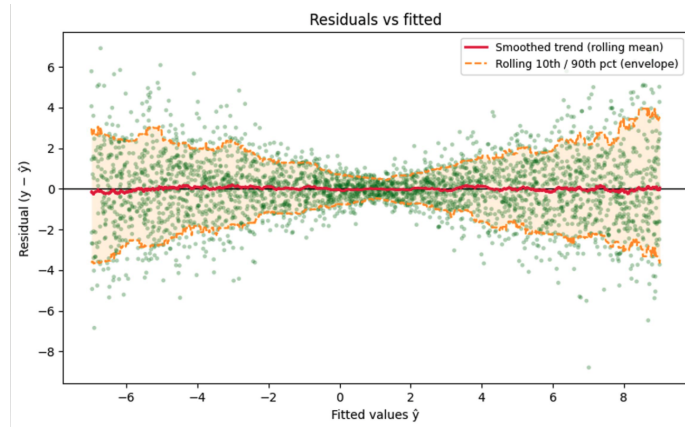
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May 2026

Questions

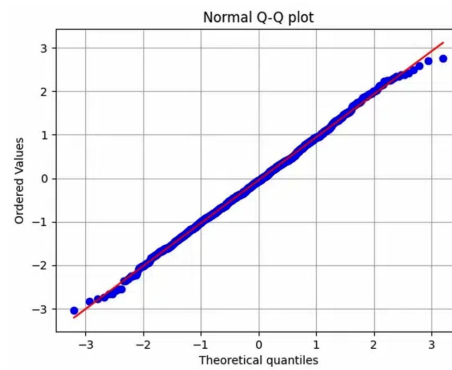
1. The estimate $\hat{\beta}$ is a random variable with mean β
 - (a) True
 - (b) False
2. According to linear regression, the following is a linear model: $\hat{Y} = \beta_0 + \beta_1 X + \beta_2 X^2$
 - (a) True
 - (b) False
3. According to linear regression, the following is a linear model: $\hat{Y} = \beta_0 + \beta_1 X + \beta_2^2 X^2$
 - (a) True
 - (b) False
4. Let's say $\alpha = 5\%$. What are we likely to conclude if we get a p-value of 0.03?
 - (a) The likelihood of observing our data under the null hypothesis is low, so we reject the null hypothesis
 - (b) The likelihood of observing our data under the null hypothesis is low, so we FAIL to reject the null hypothesis
 - (c) The likelihood of observing our data under the null hypothesis is low, so our observation is wrong, and we fail to reject the null hypothesis

5. Does the following plot indicate homoscedasticity?



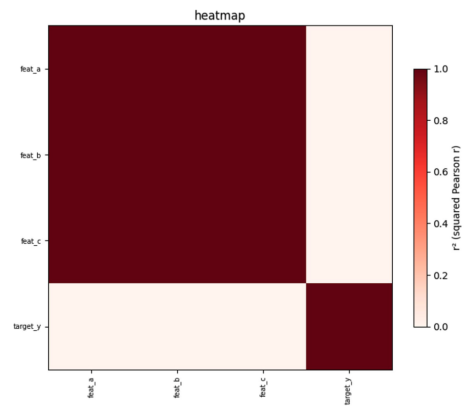
- (a) Yes
- (b) No

6. Does this NQQ plot support the normality of residuals assumption for linear regression?



- (a) Yes
- (b) No

7. The following correlation matrix is indicative of high multicollinearity



- (a) True
- (b) False

Solutions

- 1. True
- 2. True
- 3. False
- 4. (a)
- 5. No
- 6. Yes
- 7. True